Olt Lecaj

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Education

Manhattan University — Bronx, NY
Bachelor of Science in Electrical Engineering; GPA: 3.88/4.0 — *Expected May 2028*

Experience

Mission Critical PDS Intern — Jun 2026 – Aug 2026
Jones Lang LaSalle (JLL) — *Yonkers, NY*

- Produced an as-built electrical one-line diagram in AutoCAD for a 375 kW Con Edison data center, tracing power flow from two 4.16 kV utility feeds through switchgear, UPS, and PDU levels
- Authored Methods of Procedure (MOPs) for electrical equipment repair and infrared scanning, adopted by technicians and engineers for field execution
- Executed monthly generator load testing; logged and trended performance readings to verify backup-power readiness
- Supported electrical, mechanical, and thermal operations, including chiller inspections and daily critical-facility rounds

Electrical Helper (part-time) — 2022 – Present
Family Local Residential Projects — *Yonkers, Bronx, Catskill, NY*

- Assisted with residential branch-circuit wiring, pulled cable through walls/ceilings, and routed to the main panel
- Learned panel layout and circuit-breaker operation (trip/reset behavior, load paths), reinforcing core circuits coursework

Research & Projects

Hybrid-Storage DC Microgrid Testbed — Sep 2026 – Present
7x24 Exchange University Challenge — *Bronx, NY*

- Designing a bench-scale 48 V battery/supercapacitor DC bus emulating AI cluster load swings, with cooling as a controllable load; building toward a hardware-in-the-loop bench with a TI C2000 supervisory controller as device under test
- Modeled the converter in Typhoon HIL and designed a two-loop controller (fast current loop inside a slower voltage loop), verifying in simulation that the bus voltage holds steady during sudden load changes
- Specified the supervisory strategy from a HESS literature survey, catching that the published architecture omits SOC feedback and would collapse the supercapacitor; adding SOC recovery and feedforward to cut peak battery current and dI/dt

Fault Localization and Classification Using ML in Power Distribution Systems — Jun 2026 – Present
Jaspers Summer Research Scholars — *Bronx, NY*

- Ran short-circuit simulations (LL, LG, LLG, and three-phase) on the EPRI Ckt7 distribution feeder in OpenDSS, then built an ML pipeline (Random Forest, SVM) to localize faults and classify fault type
- Engineered features from PMU and smart-meter data (symmetrical components, voltage/current magnitudes and angles) plus a distance regressor for fault location
- Modeled realistic grid conditions with varying Distributed Generation penetration and smart-meter dropout rates to address bidirectional current flow and sparse-data challenges in utility protection schemes

Leadership & Honor Societies

President — Sep 2026 – Present
7x24 Exchange Student Chapter at Manhattan University — *Bronx, NY*

- Coordinating guest speakers, conferences, and networking events
- Organized an NFPA 70E electrical safety training for all ME and EE students; participants received a certificate of completion
- Leading the University Challenge team building a supercapacitor/battery controller to reduce battery degradation and grid instability

Vice President — Sep 2026 – Present
IEEE — *Bronx, NY*

- Coordinating events, scholarship opportunities, and conferences for student club members

Eta Kappa Nu | *Member; top quarter of junior EE students; tutor students 3 hrs/week* — Jan 2026 – Present

Tau Beta Pi | *Member; top eighth of junior engineering students* — Jan 2026 – Present

James Patterson Honors Program | *Mentor to younger students on college life, careers, and coursework* — Sep 2026 – Present

Technical Skills

Programming: C, C++, Python, MATLAB
Power/EE Tools: AutoCAD, OpenDSS, Typhoon HIL, oscilloscopes, multimeters, Microsoft Excel, PowerPoint
Power Systems: Power distribution architecture, load paths, load flow, power electronics (rectifiers, inverters, UPS topologies), short-circuit analysis
Certifications/Training: QEW202 Arc Flash; NFPA 70E Electrical Safety; FE exam scheduled for June 2027
Coursework: Fundamentals of Electrical Systems I/II, Signals and Systems I/II, Electronic Systems I/II, Computer Architecture, Embedded Systems, Electromagnetics, Instrumentation Systems